ZEDPATH

Team Kestrel · IntelliCon '26 Buildathon

ZP-DOC-03

Software Requirements Specification

Functional, non-functional and business-rule requirements for ZedPath Release 1 and 2

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1.0	5 Oct 2026	B.M.P Banneka	Approved by the product owner; baselined
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2.0	14 Oct 2026	B.M.P Banneka	Approved by the product owner; baselined. Supersedes 1.0

Version	Date	Author	Description of change
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Review and approval

This document takes effect when the approver signs below. Until then its status is as shown on the cover.

Role	Name	Signature	Date
Reviewer	Claude (AI coding partner), consistency and traceability review		
Approver	B.M.P Banneka, Product Owner	v2.0 approved in writing (project channel) on 14 Oct 2026 and v1.0 on 5 Oct 2026; this revision awaits approval	14 Oct 2026 (v2.0); 5 Oct 2026 (v1.0)

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1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) states precisely what ZedPath **shall** do and how well it **shall** do it. Every requirement is uniquely identified, verifiable, and traced to the user story it comes from (ZP-DOC-02). Designers use it to build the system; testers use it to prove the system is correct.

1.2 Scope

The SRS covers Release R1 (Gate 3, 11 October 2026) and Release R2 (after the buildathon). Each requirement states its release. Requirements for R2 are included so the design does not have to be reworked later.

1.3 Intended audience

Developers, testers, the product owner, and buildathon judges assessing engineering quality.

1.4 Relationship to other documents

Table 1: Position of this document in the ZedPath document set

Document	Relationship
ZP-DOC-01 Vision and Scope	Grandparent. Business objectives BO-1 to BO-6 become quality requirements here
ZP-DOC-02 User Stories	Parent. Every functional requirement cites the story and acceptance criteria it refines
ZP-DOC-04 Use Case Model	Child. Use cases realise groups of functional requirements
ZP-DOC-05 Data Design	Child. Business rules and data requirements here become entities and constraints
ZP-DOC-06 Software Architecture	Child. Non-functional requirements here drive architecture decisions
ZP-DOC-08 Test Plan	Child. Every requirement here maps to at least one test case

1.5 Conventions

- Requirements use **shall** (mandatory), **should** (recommended) and **may** (optional), as defined in ZP-DOC-00.
- **FR-enn**: functional requirement (*e* = feature/epic number). **NFR-nnn**: non-functional requirement. **BR-nnn**: business rule, a fact or policy of the domain that the system must respect.
- **Priority** uses MoSCoW (M, S, C), inherited from the parent story unless stated.
- **Verification method** (ISO/IEC/IEEE 29148): **T** test, **D** demonstration, **I** inspection, **A** analysis.
- **Change markers** (since v1.0): a requirement added or changed by a change request starts with a marker in italics: *New (CR-nnn)*, *Revised (CR-nnn)*, *Superseded by X (CR-nnn)* or *Withdrawn (reason)*. A superseded or withdrawn requirement keeps its ID and text for traceability (it is *Deleted* in the sense of ZP-DOC-00) and no longer has to be met. A new requirement is in force from this revision's approval.

1.6 Change requests since version 1.0

Table 2: Change requests and the requirements they affect (the stories are listed in ZP-DOC-02)

CR	Date	Change	Requirements
CR-001	5 October 2026	Student accounts, key derived on the phone, recovery code, cross-device profile, delete account	FR-104, FR-114, FR-115, FR-117 to FR-119, NFR-036, NFR-037, NFR-075, BR-050, BR-051, BR-058
CR-002	11 October 2026	Journey with reminders on the phone; Alerts checked on the phone; Groups with verified seniors, moderation, report, block and rate limits; new navigation	UI-1, UI-5, SI-4, SI-8, SI-9, FR-401, FR-402, FR-405, FR-406, FR-412, FR-503, FR-509, FR-510, FR-703 to FR-705, FR-707 to FR-711, NFR-039, NFR-072, BR-070 to BR-073, BR-076
CR-003	11 October 2026	Account required at the start with name, email, Sri Lankan mobile and date of birth, purpose stated and consent recorded	FR-101, FR-116, FR-120, FR-121, NFR-030, NFR-070, BR-052 to BR-057
CR-004	11 October 2026	Ask answer card: verdict, layout, actions, sources, follow-ups, feedback	FR-606 to FR-611, NFR-073
VR	11 October 2026	Video review: languages, search, Compare, chart values, Ask context, reminder settings, resit path removed, multi-select	FR-123, FR-124, FR-213, FR-215 to FR-218, FR-309, FR-310, FR-413, FR-501, FR-508, FR-612, NFR-074, BR-063, BR-080
CR-005	11 October 2026	Group rules before taking part; no phone numbers or emails in messages; popular groups; labelled samples; chat layout	UI-4, FR-712, FR-713, FR-715, FR-716, BR-074
CR-006	11 October 2026	Rules once per account; group info page; preset avatars; rooms stay connected; selection on My list with Compare; Journey ticked by the student within windows; follow courses and notices; Journey on the account; Alerts into the Journey	FR-122, FR-311, FR-401, FR-408, FR-409 to FR-411, FR-414 to FR-417, FR-713, FR-714, FR-717 to FR-719, NFR-038, NFR-043, NFR-071, BR-060 to BR-062, BR-075, BR-077

1.7 Change requests since version 2.0

New in v2.1 (draft). Requirements added after v2.0 was approved on 14 October 2026. "VR2" is the product owner's review and decisions of 14 October 2026. "Audit run-1" is the first security audit of 15 October 2026, reported in ZP-DOC-08.

Table 3: Change requests since version 2.0 and the requirements they affect

CR	Date	Change	Requirements
VR2	14 October 2026	Bands shown as Very likely, Possible and Unlikely with a one-line meaning; an expired session returns to *Log in*	FR-205 (revised), FR-221, FR-129, BR-037
CR-007	15 October 2026	*Try the demo*: private throwaway account with the sample student, deleted after 24 hours; read-only in groups	FR-125 to FR-127, BR-059
CR-008	15 October 2026	University facts from the UGC report *Sri Lanka University Statistics 2024* on the course page	FR-219, NFR-007, NFR-055, BR-081
CR-009	15 October 2026	Aptitude tests on the course page; NVQ 5/6 to NVQ 7 ladder on route R042	FR-220, FR-511, BR-064

CR	Date	Change	Requirements
CR-010	15 October 2026	Security hardening from audit run-1 (leads 1 to 8 and 10 to 17) and the test campaign (D-03, D-04, D-05, D-12)	FR-126, FR-128, FR-130 to FR-132, FR-613 to FR-615, FR-720 to FR-723, FR-914, NFR-054, NFR-076 to NFR-083, BR-059, BR-078, BR-079

2 Overall description

2.1 Product perspective

ZedPath is a new, self-contained web application. It is not part of, and does not connect to, any UGC or government system: it **reads** their public documents and helps students act on them. The system context is shown in ZP-DOC-01. It runs entirely on the Cloudflare free plan (ZP-DOC-06).

2.2 Product functions

At the highest level, ZedPath shall let a student: create an account with consent and enter their results once (FE-1); see every reachable state-university course with a likelihood band and reasons (FE-2); build and check a UGC preference list (FE-3); follow a journey they tick themselves, with followed courses and notices and reminders (FE-4); see every other route and the notices open now (FE-5); ask cited questions in three languages (FE-6); share with family and talk with peers and verified seniors in moderated groups (FE-7); and prepare for university (FE-8). Behind these, ZedPath shall keep every fact sourced, checked and correctable (FE-9).

2.3 User classes and characteristics

Table 4: User classes

Class	Persona	Frequency	Technical skill	Key needs
Student	P1, P2	Intense for 4 to 8 weeks after results, then occasional	Smartphone-native; limited data	Accuracy, speed, own language
Family member	P3	Occasional, through shared cards	Low	A simple summary in Sinhala or Tamil
Teacher or counsellor	P4	Weekly in application season	Moderate	Class overview
Verified senior	P5	Occasional	High	Safe, verified space
Curator	P6	Daily in season, weekly otherwise	High	Fast review of extracted facts

2.4 Operating environment

- Client: mobile web browsers (Chrome for Android 120+, Samsung Internet 24+, Safari for iOS 17+) and current desktop browsers. Screen widths from 360 px. *Revised (CR-002):* also the ZedPath Android app, which loads the same web application and posts native notifications (UI-5).
- Server: Cloudflare Workers runtime and Cloudflare data services (free plan).

2.5 Design and implementation constraints

These constraints are binding. Their detail is in ZP-DOC-06.

- **CON-1** The system shall run within the Cloudflare **free plan** limits verified on 5 October 2026.
- **CON-2** The application, API and tooling shall be written in **TypeScript**.
- **CON-3** The user interface shall follow **Material Design 3**.

- **CON-4** The system shall not send email, SMS or WhatsApp messages (not available or not free). *Revised (CR-003): this holds until a sending domain and an SMS service exist; email addresses and mobile numbers collected under FR-116 are therefore not verified yet.*
- **CON-5** Official third-party documents shall be stored privately and cited, not redistributed.

2.6 Assumptions and dependencies

As stated in ZP-DOC-01 (AS-1 to AS-4, DE-1 to DE-3).

3 External interface requirements

3.1 User interfaces

- **UI-1** *Revised (CR-002, CR-006).* The interface shall be a responsive, mobile-first web application with bottom navigation of five destinations (Paths, My list, Journey, Groups, Ask) and *Me* (the student's avatar) in the top bar, as specified in ZP-DOC-07. Alerts are the Journey's *Open now* tab; the top bar has no bell (FR-417). The v1.0 text was "at most four destinations (Journey, Courses, Routes, Ask)".
- **UI-2** Every screen shall offer a language switch between Sinhala, Tamil and English.
- **UI-3** Every displayed fact shall offer access to its source (document, page, date verified).
- **UI-4** *New (CR-005).* Group screens shall follow a chat-app layout: a list of groups, a room with message bubbles, and a message box at the bottom.
- **UI-5** *New (CR-002).* An Android app shall load the same web application in its own window and post native notifications for reminders (FR-412).

3.2 Software interfaces

Table 5: Software interfaces

ID	External system	Purpose	Protocol and format
SI-1	Google Gemini API, through Cloudflare AI Gateway	Read documents and photos; generate answers in three languages	HTTPS, JSON
SI-2	Cloudflare Workers AI	Embeddings and fallback inference	Workers binding
SI-3	Government Gazette website	Fetch weekly Gazette issues	HTTPS, PDF
SI-4	Web Push services of browsers. *Superseded by SI-8 (CR-002).*	Deliver reminders	Web Push protocol (RFC 8030) with VAPID (RFC 8292)
SI-5	Device calendar	Import deadlines	iCalendar file (RFC 5545)
SI-6	Device share sheet	Share the family card	Web Share API
SI-7	Google Sign-In (R2)	Verify university accounts of seniors	OpenID Connect
SI-8	Android notification system, through the Android app. *New (CR-002).*	Show reminders scheduled on the phone	Local notifications (Capacitor plugin)
SI-9	Cloudflare Workers AI moderation model. *New (CR-002).*	Screen each group message before others see it	Workers binding

3.3 Communications interfaces

- **CI-1** All traffic shall use HTTPS (TLS 1.2 or later).
- **CI-2** The public API shall exchange JSON (UTF-8).

4 Functional requirements

Each table lists the requirements of one feature. *Source* names the parent story and acceptance criteria.

4.1 FE-1 Profile and results

Table 6: Functional requirements, FE-1 Profile and results

ID	Requirement	Source	Pri	Ver
FR-101	*Revised (CR-003): "without an account" removed; an account comes first (FR-116).* The system shall let a student enter stream, district, Z-score and three subjects with grades.	US-101, AC-101.1	M	T
FR-102	The system shall reject a Z-score outside −4.0000 to +4.0000 or with more than four decimal places, with an inline message.	AC-101.2	M	T
FR-103	The system shall reject a subject combination that is not admissible for the selected stream (BR-010) and name the offending subject.	AC-101.3	M	T
FR-104	*Superseded by FR-117 (CR-001, CR-003).* The system shall store the profile only on the student's device and restore it on the next visit.	AC-101.4	M	T
FR-105	The system shall recalculate every result immediately when any profile field changes.	AC-102.1	M	T
FR-106	The system shall delete the profile from the device on request.	AC-102.2	M	T
FR-107	The system shall estimate a Z-score **range** from expected marks, using at least the three most recent years of subject statistics, labelled *Estimate*.	US-103	S	T
FR-108	When an estimated range is used, bands shall be computed from the lower bound and marked as estimate-based.	AC-103.3	S	T
FR-109	The system shall read stream, subjects, grades, Z-score and district from a results-sheet photo and require the student to confirm each field before use.	US-104	S	D
FR-110	The system shall highlight every field read with confidence below 0.85 for the student to check.	AC-104.2	S	T
FR-111	The system shall delete an uploaded photo as soon as reading finishes, and record the deletion in the audit log.	AC-104.3	S	I
FR-112	The system shall switch all interface text between Sinhala, Tamil and English without losing the current screen or data.	US-105	M	D
FR-113	When an offering has an O/L requirement (BR-022), the system shall ask for the relevant O/L grades, evaluate them, and show the offering as *Check O/L requirement* until they are given.	US-201; BR-022	M	T

ID	Requirement	Source	Pri	Ver
FR-114	*New (CR-001).* The system shall create a student account from a username (BR-050) and a password (BR-051). The device shall derive a key from the password with PBKDF2-HMAC-SHA256 at 600,000 iterations and send only that key; the password shall never be sent (NFR-036).	US-106, AC-106.2	M	T
FR-115	*New (CR-001).* The system shall let a student log in with username and password on any device, keep the session for 30 days (BR-058), and log out on this device or on every device.	US-106	M	T
FR-116	*New (CR-003).* The system shall require an account before any screen other than the welcome, sign-up and log-in screens. Sign-up shall require full name, email address, Sri Lankan mobile number and date of birth (BR-052 to BR-055), shall state the purpose of each before consent, and shall be refused by the server when any of them, or the consent, is missing or invalid.	US-106, AC-106.1; US-109, AC-109.1 to AC-109.3	M	T
FR-117	*New (CR-001; supersedes FR-104).* The system shall keep the student's profile (results, achievements, interests) on the account, restore it on log-in on any device, and keep a copy on the device. A change made offline shall be sent when the device is online and shall not be overwritten by the older copy on the server.	US-106, AC-106.3, AC-106.4; AC-101.5	M	T
FR-118	*New (CR-001).* The system shall show an 80-bit recovery code once at sign-up and require the student to confirm saving it; it shall let the student set a new password with username and recovery code, then issue a new code and invalidate the old one. Typed 0, 1 and 8 shall be read as O, I and B.	US-107	S	T
FR-119	*New (CR-001).* On confirmed request, the system shall delete the account and every row stored under it (NFR-075), delete the account's messages from every group it was in, clear the student's data from the device and return to the welcome screen.	US-108	M	T
FR-120	*New (CR-003).* An account made before CR-003 without contact details shall be asked to complete them before any other screen; no value shall be invented for it.	AC-109.4	M	T
FR-121	*New (CR-003).* The consent statement shall include that a parent or guardian agrees when the student is under 18 (BR-056); the system shall record the consent version and time (BR-057), and shall tell the student that email and mobile are not verified yet (CON-4).	AC-109.2, AC-109.5	M	T
FR-122	*New (CR-006).* The system shall let a student choose one of 12 preset avatars, or none (initials, the default), shall accept no other value and no photo upload (BR-077), and shall show the avatar in the app bar, in group member lists and with the group messages the student posts.	US-110	C	T

ID	Requirement	Source	Pri	Ver
FR-123	*New (VR).* The system shall filter the subject list in *About you* as the student types: every word typed must start a word of the subject name, common short forms are understood, and case, accents and punctuation are ignored.	US-111	S	T
FR-124	*New (VR).* While the Sinhala and Tamil interface text has not been reviewed by native speakers, the system shall say that it is an AI translation being checked; official names and quotations shall stay in English as published.	AC-105.3, AC-105.4	M	I
FR-125	*New (CR-007).* On *Try the demo*, the server shall create a demo account holding the sample student (profile, achievements, interests and sample contact details, accepted by the same rules as a sign-up), start a 24-hour session and record the account's expiry (BR-059). The account shall have a random username and an unusable random password verifier, so nobody can log in to it with a password.	US-112, AC-112.1 to AC-112.3	S	T
FR-126	*New (CR-007; bounded by CR-010).* Each demo tap shall first delete up to 50 demo accounts whose expiry has passed, with every row stored under them, and no other account; a failed clean-up shall not block the tap. When 500 demo accounts exist, the tap shall be refused with a message inviting the visitor to create an account (BR-059).	AC-112.4, AC-112.7	S	T
FR-127	*New (CR-007).* A demo account shall be able to join and read a group that already exists; the server shall refuse it creating a group or reporting a message (403), and the group room shall refuse its posts.	AC-112.5	S	T
FR-128	*New (CR-010).* Demo mode shall prefill no password, and no password shall be shipped in the app or its scripts. Clearing the phone (*Delete everything on this phone* or *Log out and clear this phone*) shall also switch demo mode off.	AC-102.3, AC-112.6	S	T
FR-129	*New (VR2).* When the server answers 401 to any request outside the sign-in routes (sign-up, log-in, recovery, demo, log-out), the app shall end the local session and open *Log in*.	AC-106.5	S	D
FR-130	*New (CR-010).* The app shall record on the phone which username its stored results, list, Journey and settings belong to; when a different student logs in, it shall clear that data from the phone first (the language choice stays). Clearing the phone shall also forget the owner.	AC-102.4	M	T
FR-131	*New (CR-010).* When *Log out* or *Log out everywhere* fails, the app shall say that the student is still logged in and should try again, and shall not show the student as logged out.	AC-106.6	S	D

ID	Requirement	Source	Pri	Ver
FR-132	*New (CR-010).* The server shall record each group room an account opens (GROUP_ROOM_SEEN, migration 0012, ZP-DOC-05); deleting the account shall purge the account's messages from every room it is in or ever opened, read before the rows are deleted.	AC-108.3	M	T

4.2 FE-2 State-university course discovery

Table 7: Functional requirements, FE-2 State-university course discovery

ID	Requirement	Source	Pri	Ver
FR-201	The system shall list every course offering for which the student meets the general eligibility rules (BR-001 to BR-004) and the course's specific subject and grade requirements (BR-020 to BR-024), excluding suspended offerings (BR-026).	US-201	M	T
FR-202	The eligibility check shall evaluate alternative requirements ("A **or** B") and combined requirements ("A **and** B") exactly as stated in the handbook.	AC-201.3	M	T
FR-203	The eligibility check shall run on the server; the client may cache results but shall not be the only place the rules are applied.	US-201; DoD 2	M	I
FR-204	The system shall show the number of hidden offerings and, for each, the rule failed in plain language with a link to the handbook page.	US-202	M	T
FR-205	*Revised (VR2): the names shown to students changed (FR-221); the bands and their rules did not.* The system shall assign each eligible offering exactly one band (Safe, Likely, Reach, Out of range or Not enough data) using the banding rules BR-030 to BR-036.	US-203	M	T
FR-206	For each banded offering, the system shall show the signed difference between the student's Z-score and the latest cut-off (four decimals) and the trend (rising, steady, jumpy).	AC-203.3	M	T
FR-207	No screen or answer shall state or imply that admission is certain.	AC-203.4	M	I
FR-208	The system shall show, per offering, the cut-off for the student's district for each available intake year, with the student's Z-score as the last row.	US-204	M	T
FR-209	Every cut-off, rule, duration, medium and aptitude-test flag shown shall link to its source document and page and show the date verified.	AC-204.2	M	I
FR-210	The system shall filter offerings by band, university, field of study and distance from the student's district, and update the count.	US-205	S	T
FR-211	For a chosen dream offering, the system shall show the gap to its lowest cut-off in the available years and classify the gap as *small* (≤ 0.10) or *large* (> 0.10).	US-206	S	T
FR-212	For a dream offering, the system shall suggest up to three offerings in the same field that are Safe or Likely for the student.	AC-206.2	S	T

ID	Requirement	Source	Pri	Ver
FR-213	*Revised (VR): "two" became "up to three".* The system shall compare up to three offerings side by side on band, duration, medium, campus, aptitude test and trend.	US-207, AC-207.1, AC-207.2	C	D
FR-214	The system shall show monthly living-cost estimates by campus with sample size and collection date.	US-208	C (R2)	D
FR-215	*New (VR).* The system shall search the student's courses across every band by course name, university name, common short forms or Uni-Code, with the matching rule of FR-123.	US-209	S	T
FR-216	*New (VR).* Compare shall show each offering as a column that swipes sideways on a narrow screen and, while fewer than three are compared, a "+" column that opens a picker of the student's Safe, Likely and Reach courses (searchable) and of My list.	AC-207.2, AC-207.3	C	D
FR-217	*New (VR).* Every *Compare* button in the app shall open the Compare screen with that offering included.	AC-207.4	C	D
FR-218	*New (VR).* The cut-off chart on a course page shall print each point's cut-off value beside the point.	AC-204.3	S	I
FR-219	*New (CR-008).* The course page shall show *University facts* for the offering's institution from the UGC report *Sri Lanka University Statistics 2024*: students admitted for 2023/24 (Table 2.2), undergraduates in 2024 (Table 3.1(a)), permanent hostel places and the share of undergraduates they can house (Table 7.1), Mahapola and Bursary recipients processed in 2024 (Table 7.2) and graduates of 2024 (Table 4.1), each with its table, page and a link to that page of the official PDF (BR-081, NFR-055).	US-210	C	T
FR-220	*New (CR-009).* For each of the 24 offerings with an aptitude or practical test (BR-025), the course page shall show the test's institution, application opening and closing dates and test date from its official notice, with a link to the notice; dates shall be labelled as the last intake's and past under BR-064. A course whose joint practical test is held at other universities shall link the official notices of those universities. No card shall be shown for a course without a test.	US-211	S	T
FR-221	*New (VR2).* The interface shall name the bands *Very likely* (SAFE), *Possible* (LIKELY) and *Unlikely* (REACH) in all three languages, and show the one-line meaning of the selected band under the *Courses* tabs (BR-037).	AC-203.5	M	I

4.3 FE-3 Preference list

Table 8: Functional requirements, FE-3 Preference list

ID	Requirement	Source	Pri	Ver
FR-301	The system shall let the student add eligible offerings to an ordered list and reorder them by drag and drop and by keyboard.	US-301	M	T
FR-302	The system shall refuse to add an offering for which the student is not eligible.	AC-301.3	M	T
FR-303	The system shall warn when a Safe offering is listed above a Likely or Reach offering (BR-040, BR-041).	US-302	M	T
FR-304	The system shall warn when the list contains no Safe offering (BR-042).	AC-302.2	M	T

ID	Requirement	Source	Pri	Ver
FR-308	The system shall limit the list to 125 Uni-Codes with no duplicates (BR-043).	US-301; BR-017	M	T
FR-305	The system shall propose a corrected order that keeps the student's most-wanted offerings first and Safe ones last, applied only after the student confirms.	AC-302.3	M	T
FR-306	The system shall copy the list's uni-codes to the clipboard, one per line, in list order.	US-303	M	T
FR-307	The system shall present the official application steps with the most common mistake for each, linked to the official instructions.	US-304	S	I
FR-309	*New (VR, CR-006).* Pressing and holding a course in *Courses* or *My list*, or tapping *Select*, shall start selection with that course ticked; while selecting, a tap shall tick or untick a course instead of opening it.	US-305, AC-305.1	C	T
FR-310	*New (VR).* The selection bar in *Courses* shall offer *Add to my list*, which adds every selected course under FR-302 and FR-308 and reports how many were added, how many were already listed, and whether the list was full.	AC-305.2	C	T
FR-311	*New (CR-006).* The selection bar in *Courses* and *My list* shall offer *Compare*, which opens Compare with the selected courses in the order selected, at most three.	AC-305.3	C	T

4.4 FE-4 Journey and deadlines

Table 9: Functional requirements, FE-4 Journey and deadlines

ID	Requirement	Source	Pri	Ver
FR-401	*Revised (CR-006): past stages are no longer marked done by date (FR-409).* The system shall show the student's Journey steps in order (BR-060), the courses and notices the student follows, and the next dated item at the top with days remaining.	US-401, AC-401.2	M	T
FR-402	*Revised (CR-002, CR-006): no estimated dates.* A date shall be shown only when it is announced, labelled *Confirmed* with its source. A UGC step whose date is not announced shall show *Not announced yet* with the official rule and its handbook page; the system shall never invent or estimate a date.	US-402, AC-402.1, AC-402.3	M	T
FR-403	When the list contains an offering with an aptitude or practical test (BR-025), the journey shall add the test's application and test dates.	US-403	M	T
FR-404	*Revised (VR): alarms follow the reminder settings (FR-413).* The system shall generate an iCalendar file of the journey's dated items with alarms 7 days and 1 day before each, as set in FR-413.	AC-404.1	S	T
FR-405	*Superseded by FR-412 (CR-002).* The system shall send web-push reminders 7 days and 1 day before each deadline to students who opted in.	AC-404.2	S	T
FR-406	*Superseded by FR-412 (CR-002): no push subscription is stored on the server.* The system shall delete a push subscription when the student turns notifications off.	AC-404.3	S	T

ID	Requirement	Source	Pri	Ver
FR-407	The system shall check special-intake categories against the student's answers and link each to its handbook section.	US-405	C (R2)	T
FR-408	*Superseded by FR-414 and FR-416 (CR-006).* The system shall let the student add a route's dates to the journey.	US-406	C (R2)	D
FR-409	*New (CR-006).* Every Journey step shall start unticked; only the student shall tick or untick a step.	US-407, AC-407.1	M	T
FR-410	*New (CR-006).* The server shall accept a tick only inside the step's window (BR-061) and shall refuse a tick outside it without storing anything; the app shall disable the tick box and state why.	AC-407.2 to AC-407.4	M	T
FR-411	*New (CR-006).* The system shall save the student's followed courses, followed notices (with the *I applied* state) and ticked steps on the account, show the same Journey on every device, and delete it with the account.	US-411	M	T
FR-412	*New (CR-002; supersedes FR-405, FR-406).* In the Android app, the phone shall schedule a notification 7 days and 1 day before each dated item in the Journey, at the hour set in FR-413; no push subscription or device identifier shall be stored on the server for reminders. On other devices the calendar file (FR-404) provides the reminders.	AC-404.5, AC-404.6	S	T
FR-413	*New (VR). Me* shall offer reminder settings, kept on the phone: 7-day reminder on or off, 1-day reminder on or off, 8 am or 6 pm, and the new-notice count on or off. Unset or unreadable settings shall fall back to the default (both reminders on, 8 am, count shown), field by field (BR-063). There shall be no test-notification button.	US-408	S	T
FR-414	*New (CR-006; with FR-416 supersedes FR-408).* The student shall be able to follow and unfollow a notice or other path from *Open now* and from other-path pages; a followed one shall appear in the Journey with its closing date and source. The server shall accept only identifiers in ZedPath's route catalogue.	US-410, AC-410.1, AC-410.4	S	T
FR-415	*New (CR-006).* The student shall be able to tick *I applied* on a followed notice only once it has opened (BR-062); the server shall refuse it before then, and for a notice that is not followed.	AC-410.2, AC-410.3	S	T
FR-416	*New (CR-006).* The student shall be able to follow and unfollow a course from its course page; the server shall accept only 2025/26 offerings.	US-409	S	T
FR-417	*New (CR-006).* The Journey shall have a second tab, *Open now*, listing the open notices (FR-503); the top bar shall have no bell, and the count of new notices shall show on the Journey tab when the setting is on (FR-413).	AC-503.5	M	D

4.5 FE-5 Other routes

Table 10: Functional requirements, FE-5 Other routes

ID	Requirement	Source	Pri	Ver
FR-501	*Revised (VR): the retry group ("Sit the A/L again") was removed.* The system shall list non-UGC routes in six groups (private and non-state degrees, higher national diplomas, professional qualifications, vocational courses, job examinations, study abroad) and mark each as open or not open to the student with the reason.	US-501, AC-501.2, AC-501.3	M	T
FR-502	Each route shall show entry requirements, duration, approximate cost, intake timing and sources with verified dates.	US-502	M	I
FR-503	*Revised (CR-002, CR-006): intake notices added; shown under the Journey's Open now tab (FR-417).* The system shall show announced Gazette job examinations and intakes of other routes, each with its closing date, provider and a link to the official notice, and the result of the eligibility check of FR-509.	US-503, AC-503.1, AC-503.3	M	T
FR-504	A job-examination notice extracted with low confidence shall not be shown until a curator approves it.	AC-503.2	M	T
FR-505	The system shall show the recognition status of a programme's awarding university and local approval, each with source and date checked.	US-504	S (R2)	I
FR-506	The system shall check the basic conditions of the government interest-free loan scheme.	US-505	S (R2)	T
FR-507	The system shall list government scholarships abroad open to school leavers with usual timing and requirements.	US-506	C (R2)	I
FR-508	*Withdrawn (VR, 11 October 2026: the "Sit the A/L again" path was removed).* For a dream offering, the system shall show the Z-score gap, the student's attempt number out of the maximum (BR-004) and alternative routes.	US-507	S	T
FR-509	*New (CR-002, CR-003).* For each notice, the device shall check the A/L-pass rule against the student's results and the age rule against the date of birth given at registration, showing each as met, not met or unknown (data missing); O/L rules shall be shown for the student to check and not evaluated. The check shall run on the device (NFR-072).	AC-503.3	M	T
FR-510	*New (CR-002).* Notices shall be ordered with open and upcoming ones first, soonest closing date first, then closed ones, most recently closed first.	AC-503.4	S	T
FR-511	*New (CR-009).* The University of Vocational Technology route (R042) shall quote the TVEC statement of the NVQ Level 5/6 to NVQ Level 7 (degree) progression, with the manual and page, and list the degree programmes the university lists for 2026 with the date they were read.	US-508	C	I

4.6 FE-6 Ask

Table 11: Functional requirements, FE-6 Ask

ID	Requirement	Source	Pri	Ver
FR-601	The system shall answer a question in the language it was asked (Sinhala, Tamil or English).	US-601	M	T
FR-602	Answers about eligibility shall use the student's profile when the student has one.	AC-601.2	M	T
FR-603	Every factual answer shall cite at least one stored source (document and page).	US-602	M	T
FR-604	When no stored source supports an answer, the system shall say it does not know and point to an official contact, and shall not generate an unsupported answer.	AC-602.2	M	T
FR-605	The system shall decline to predict future cut-offs or guarantee admission, and offer the historical pattern instead.	US-603	S	T
FR-606	*Revised (CR-004): moved to R1 and detailed.* The system shall offer actions under an answer, built by ZedPath from its own data and never from the model's text: for up to three course places, best chance first, open the course, add to list (only when eligible) and compare; for a matching notice, add to the Journey.	US-604	C	D
FR-607	*New (CR-004).* When the model's reply starts with a verdict, the answer shall show it as a chip: *Yes*, *No*, *It depends* or *Not in official sources*; with no verdict, no chip shall be shown.	AC-605.1	S	T
FR-608	*New (CR-004).* An answer shall be shown as one lead line and short bullets; bullets after the fourth shall be folded behind *Show more*.	AC-605.2	S	T
FR-609	*New (CR-004, VR).* Each citation chip shall show a readable source name (handbook page, or the route's name; never an internal file note), open the quoted passage, and link to the official document over HTTPS; a route citation shall also open the route's page in ZedPath.	AC-605.3	S	T
FR-610	*New (CR-004).* An answer about a course shall offer two follow-up questions about that course; an answer whose verdict is *Not in official sources* shall offer none.	AC-605.4	C	D
FR-611	*New (CR-004).* The system shall record *Helpful* and *Report a wrong answer* (optional note of up to 500 characters) with the question, the answer, the cited pages and the app language, and nothing that identifies the student (NFR-073); the screen shall say that the question and answer will be sent before the report is sent; submissions shall be rate-limited.	US-606	S	T
FR-612	*New (VR).* The app shall send the student's achievements and interests with a question; the server shall use them for that answer only and not store them (NFR-074). Whether the special intake may apply shall be decided by BR-080 in code and given to the model as a fact, and the special-intake handbook section shall be retrieved with the question.	US-607	C	T
FR-613	*New (CR-010).* A link that opens Ask with a question shall only fill the question box; the question shall be sent only when the student taps send (NFR-079).	AC-601.3	M	I
FR-614	*New (CR-010).* The service shall store at most 2,000 answer ratings and reports a day, answering "enough reports for today" beyond that, and shall delete those older than one year.	US-606; AC-907.3	S	T
FR-615	*New (CR-010).* Ask shall answer at most 3,000 questions a day across the whole service (the sum of the day's per-device counters), in addition to the per-device and per-network limits (NFR-034).	AC-907.3	S	T

4.7 FE-7 People around the student

Table 12: Functional requirements, FE-7 People around the student

ID	Requirement	Source	Pri	Ver
FR-701	The system shall generate a share card (image and short text) in the chosen language and pass it to the device share sheet.	US-701	M	D
FR-702	The share card shall exclude the student's name and Z-score unless the student includes them.	AC-701.2	M	T
FR-703	*Revised (CR-002): moved to R1; seniors answer in groups.* The system shall show seniors' messages with the badge "Verified senior · *course*, year *n*", set by the server from the redeemed invite and never by the member.	US-702	S	D
FR-704	*Superseded by FR-707 (CR-002).* The system shall verify a senior by a university Google account on a recognised domain, or by curator review of proof that is deleted after the decision.	US-703	S (R2)	T
FR-705	*Revised (CR-002): moved to R1 and detailed.* The system shall provide one group per target course and one per A/L batch (BR-070, BR-071); *Groups* shall list the groups for the courses on the student's list (or the three best-chance courses when the list is empty), the student's batch group and every joined group; only members shall open a group's live room.	US-704, AC-704.1 to AC-704.4	C	T
FR-706	The system shall show a teacher aggregate counts for students who shared with the teacher's class code, and names only of those who shared.	US-705	C (R2)	T
FR-707	*New (CR-002; supersedes FR-704).* The system shall verify a senior by a one-time invite code issued by the product owner after checking enrolment, naming course, institution and year of study (BR-076). Redeeming it at sign-up shall create a senior account that keeps no A/L results; a used or expired code shall be refused.	US-703, AC-703.4, AC-703.5	S	T
FR-708	*New (CR-002).* Group messages shall be text only, 1 to 500 characters, and shall be checked by an automatic moderation model before any other member sees them; a message judged unsafe, or one the check cannot judge, shall not be posted (NFR-039), and the sender shall be told why.	AC-706.1	M	T
FR-709	*New (CR-002).* The room shall limit each member to one message every 3 seconds and 20 messages in any 10 minutes (BR-072).	AC-706.2	M	T
FR-710	*New (CR-002).* A member shall be able to report a message once, with a reason (BR-073); a message reported by three different members shall be hidden for everyone.	AC-706.3	M	T
FR-711	*New (CR-002).* A member shall be able to block and unblock another member by username; a blocked member's messages shall not be shown or delivered to the blocker.	AC-706.4	M	T
FR-712	*New (CR-005).* The room shall refuse a message containing a Sri Lankan phone number or an email address, with an explanation (group rule 3, BR-074).	AC-706.5	M	T

ID	Requirement	Source	Pri	Ver
FR-713	*Superseded by FR-714 (CR-006).* A member shall accept the current group rules for each group before taking part; the server shall refuse a join without the current rules version and keep the live room closed for a member whose membership is not on it.	AC-707.1	M	T
FR-714	*New (CR-006; supersedes FR-713).* A student shall accept the current group rules once per account, at the first join of any group (BR-075); the server shall record the version and time, refuse a first join without acceptance, and keep every live room closed until the account has accepted the current version. A new rules version shall be asked for again.	AC-707.2 to AC-707.5	M	T
FR-715	*New (CR-005). Groups* shall list popular course groups for everyone: Medicine (001), Engineering (008), Computer Science (012), Management (016), Commerce (018) and Law (025).	AC-704.5	C	D
FR-716	*New (CR-005).* A group may show curated sample conversations before the real messages; each shall be marked *Sample*, have an author name starting "Sample" and no verified badge, shall not be reportable or blockable, and shall not be stored as a message.	AC-704.6	C	T
FR-717	*New (CR-006).* Tapping the group's name in a room shall open its info page: description, general group rules and any group-specific rules, the member list, and *Exit group* at the bottom. The (i) button shall be removed.	US-708, AC-708.1	S	D
FR-718	*New (CR-006).* The member list shall show usernames, avatars and senior badges, never real names, and the server shall return it only to members of the group (NFR-071). *Exit group* shall end the membership and close the room for that student.	AC-708.2, AC-708.3	S	T
FR-719	*New (CR-006).* An open group room shall be kept alive by a message from the app at least every 25 seconds and shall reconnect by itself after a dropped connection, without showing "Disconnected" (NFR-043).	AC-704.8	S	T
FR-720	*New (CR-010).* Only reports from accounts at least one day old that are not demo accounts shall count toward the three that hide a message (BR-078); other reports shall still be recorded.	AC-706.6	M	T
FR-721	*New (CR-010).* The room shall count every post attempt, refused or not, before moderation, and refuse further attempts after 20 in any 10 minutes per member (BR-079).	AC-706.7	M	I
FR-722	*New (CR-010).* A room shall keep at most three live connections per account, closing the oldest; opening the live room shall be rate-limited per account (BR-079).	AC-706.7	M	I
FR-723	*New (CR-010; test-campaign finding D-04).* The contact check of FR-712 shall also run on a normalised copy of the message (Unicode NFKC, invisible characters removed, spelled-out digits turned into digits) and shall refuse 0094 and other loose phone forms, "[at]" and "(dot)" email forms, and WhatsApp or Telegram invite links, while allowing years, index numbers and Z-scores.	AC-706.8	M	T

4.8 FE-8 After selection

Table 13: Functional requirements, FE-8 After selection (Release R2)

ID	Requirement	Source	Pri	Ver
FR-801	The system shall present each university's registration requirements as a checklist with where to obtain each item.	US-801	C	I
FR-802	The system shall explain Mahapola and bursary eligibility and the documents that delay applications.	US-802	C	I
FR-803	The system shall propose a month-by-month plan for the period before lectures, based on the course and district.	US-803	C	D
FR-804	The system shall keep a documents checklist covering every route.	US-804	C	D

4.9 FE-9 Data and trust

Table 14: Functional requirements, FE-9 Data and trust

ID	Requirement	Source	Pri	Ver
FR-901	The system shall let a curator register a source document with publisher, title, edition, issue date and language, store it unchanged, and record its SHA-256 fingerprint.	US-901	M	T
FR-902	The system shall detect a duplicate source by fingerprint.	AC-901.2	M	T
FR-903	The system shall extract, from a handbook source, one candidate rule per course with a confidence score and source page.	US-902	M	T
FR-904	An independent second model shall mark each candidate rule *agree* or *disagree*.	AC-902.2	M	T
FR-905	A candidate rule shall be published automatically only when both models agree and both confidences are at least 0.90; otherwise it shall enter the review queue.	AC-902.3	M	T
FR-906	The review queue shall show the source text, the proposed structured fact, both verdicts and the page, and let the curator approve, correct or reject.	US-903	M	D
FR-907	Each review decision shall be recorded with reviewer, time and the change made.	AC-903.2	M	T
FR-908	The system shall fetch new Gazette issues weekly, extract exam name, eligibility, age limit and closing date with confidence, and log counts per run.	US-904	M	T
FR-909	A failed scheduled run shall be retried automatically and its failure shown to the curator.	AC-904.2	M	T
FR-910	The system shall let anyone report a mistake on any fact with an optional note, linked to that fact, and queue it for the curator.	US-905	M	T
FR-911	The system shall keep every previous version of a published fact.	US-906	S	T
FR-912	The system shall require a bot check before accepting a report, a reminder opt-in or a question.	US-907	S	T
FR-913	The system shall limit Ask requests per client per minute (NFR-034) and return a friendly message when exceeded.	AC-907.2	S	T

ID	Requirement	Source	Pri	Ver
FR-914	*New (CR-010).* When a session or membership ends, the system shall close the live group-room connections opened under it: on log out, those of that session only; on log out everywhere and on password recovery, every connection of the account; on leaving a group, the account's connections to that group (NFR-077).	US-908, AC-908.1 to AC-908.3	S	T

5 Non-functional requirements

5.1 Performance

Table 15: Performance requirements

ID	Requirement	Ver
NFR-001	The first screen shall be interactive within 3 seconds on a "Fast 3G" network profile on a mid-range Android phone.	T
NFR-002	Eligibility and banding results shall appear within 2 seconds of submitting a profile on a "Fast 3G" profile.	T
NFR-003	95% of API requests (excluding Ask) shall complete within 500 ms at the edge.	T
NFR-004	Ask answers shall begin streaming within 4 seconds in 95% of cases.	T
NFR-005	Each API request shall use less than 10 ms of CPU time (the free-plan limit).	A
NFR-006	The initial JavaScript download shall not exceed 200 KB compressed.	I
NFR-007	*New (CR-008; test-campaign finding D-05).* The API shall answer 304 Not Modified to a request whose If-None-Match carries the current ETag in its strong or weak (W/) form, so a compressed response can be revalidated without being downloaded again.	T

5.2 Capacity and cost

Table 16: Capacity and cost requirements

ID	Requirement	Ver
NFR-010	The system shall serve 10,000 students per admission cycle within the free plan (BO-1, BO-6).	A
NFR-011	Daily usage of each Cloudflare free-plan quota shall stay below 70% on a typical day and below 100% on results day, according to the budget in ZP-DOC-06.	A
NFR-012	Page navigation shall not consume Worker requests (static assets are served without invoking the Worker).	I

5.3 Usability, accessibility and localisation

Table 17: Usability, accessibility and localisation requirements

ID	Requirement	Ver
NFR-020	A first-time user shall reach a checked preference list in under 10 minutes without help (BO-3), measured with at least 5 participants.	T
NFR-021	The interface shall meet WCAG 2.2 level AA.	I
NFR-022	Touch targets shall be at least 48 × 48 dp, following Material Design 3.	I
NFR-023	All interface text shall exist in Sinhala, Tamil and English; no screen shall mix languages except for official names and quotations.	I
NFR-024	Sinhala and Tamil text shall render correctly with Noto Sans Sinhala and Noto Sans Tamil.	D
NFR-025	Layouts shall work from 360 px width without horizontal scrolling.	T

5.4 Security and privacy

Table 18: Security and privacy requirements

ID	Requirement	Ver
NFR-030	*Revised (CR-003): an account is now required, so personal data is stored by consent at sign-up rather than by optional opt-in.* The system shall store personal data on the server only for a purpose stated to the student and with the student's recorded consent (FR-121), in line with the Personal Data Protection Act No. 9 of 2022 (Sri Lanka); it shall collect no personal detail beyond those in FR-116.	I
NFR-031	Results photos shall be deleted within 5 minutes of upload, whether or not reading succeeds.	T
NFR-032	Curator functions shall require authentication; every curator action shall be logged.	T
NFR-033	Secrets (API keys) shall be stored as Cloudflare secrets and never in source code or the repository.	I
NFR-034	Ask shall accept at most 10 questions per client per minute and 100 per client per day.	T
NFR-035	All input shall be validated on the server; database access shall use bound parameters only.	I
NFR-036	*New (CR-001).* A password shall never leave the device; the server shall store only a SHA-256 hash of the key derived on the device (FR-114). Session tokens and recovery codes shall be stored only as SHA-256 hashes, and the session cookie shall be HttpOnly, Secure and SameSite=Lax; cross-site writes shall be refused.	I
NFR-037	*New (CR-001).* Sign-up and log-in attempts shall be limited to 10 a minute per network address and username, and 100 a minute per network address.	T
NFR-038	*New (CR-002, CR-005, CR-006).* Group membership, group-rules acceptance, the Journey tick windows and the *I applied* rule shall be enforced by the server; checks in the app are a convenience only. Which account acts shall come only from the session, never from the request body.	T
NFR-039	*New (CR-002).* Message moderation shall fail closed: when the moderation result cannot be read, the message shall not be posted.	T

5.5 Reliability and availability

Table 19: Reliability and availability requirements

ID	Requirement	Ver
NFR-040	Core browsing (courses, bands, list) shall remain usable if the AI services are unavailable.	T
NFR-041	Scheduled jobs shall retry failed steps at least three times with back-off.	T
NFR-042	The database shall be restorable to any point in the previous 7 days (D1 Time Travel).	D
NFR-043	*New (CR-006).* An idle group-room connection shall stay open (keep-alive at least every 25 seconds, answered without waking the room), and a dropped connection shall be re-established without the student's action; the room retries after 1, 2, 4 ... up to 15 seconds and shows "Reconnecting" only after three failed tries; "Disconnected" is shown only when the account was deleted.	T

5.6 Data quality

Table 20: Data quality requirements

ID	Requirement	Ver
NFR-050	100% of published facts shall reference a source document and page (BO-2).	A
NFR-051	Cut-off values shall be stored and compared as exact four-decimal values, not binary floating point.	I

ID	Requirement	Ver
NFR-052	Imported cut-off tables shall be verified by two independent extraction methods with zero unexplained differences before publication.	T
NFR-053	A reported mistake shall be triaged within 48 hours during the admission season.	A
NFR-054	*New (CR-010).* No URL in ZedPath's data shall point to a host or a page blocked during research (CLAUDE.md research rules). Blocked pages shall be recorded only as SHA-256 fingerprints of their normalised address, so the public repository never shows the address.	T
NFR-055	*New (CR-008).* A statistical fact shall be shown exactly as printed in the official report, with the report's caveats (provisional, incomplete, including campuses); a value the report marks "Information not submitted" shall not be shown, and never as zero.	T

5.7 Maintainability and portability

Table 21: Maintainability and portability requirements

ID	Requirement	Ver
NFR-060	Business rules (eligibility, banding thresholds) shall be data or pure functions with unit tests, not scattered through the interface.	I
NFR-061	Rule-engine code shall have at least 90% branch coverage.	A
NFR-062	A new intake year's handbook and cut-offs shall be loadable without code changes.	D
NFR-063	The front end shall run in any standards-compliant browser without installation; it may be installed as a progressive web app.	D

5.8 Privacy of accounts, groups and Ask

New (CR-001 to CR-006). These requirements make the privacy promises of the change requests verifiable.

Table 22: Privacy requirements added by the change requests

ID	Requirement	Ver
NFR-070	*New (CR-003).* Contact details (full name, email, mobile, date of birth) shall be kept in their own table, apart from results; they shall never be shown to other students, never be sold, and be deleted with the account. Email and mobile shall not be unique, so that sign-up never reveals whether an address is already registered.	I
NFR-071	*New (CR-006).* In groups the system shall show only a member's username, preset avatar and any senior badge: never a real name, contact detail or uploaded photograph.	T
NFR-072	*New (CR-002).* The eligibility of notices (FR-509) shall be computed on the student's device; no request for this check shall send the student's results or date of birth.	I
NFR-073	*New (CR-004).* Answer feedback (FR-611) shall be stored without any account, device or network identifier.	I
NFR-074	*New (VR).* Achievements and interests sent with a question (FR-612) shall be used for that answer only and shall not be stored with the question or the answer.	I
NFR-075	*New (CR-001, CR-006).* Deleting an account shall remove every row stored under it in every table (account, sign-in, sessions, recovery code, redeemed senior invite, profile, subjects, achievements, interests, contact details, group memberships, rules acceptances, reports, blocks in both directions and the Journey) and its group messages, and nothing of another account.	T

5.9 Security hardening

New (CR-010). These requirements close findings of the 20-agent test campaign (D-03, D-04, D-05, D-12) and of security audit run-1 (every lead except lead 9, see below), reported in ZP-DOC-08.

Table 23: Security-hardening requirements added by CR-010

ID	Requirement	Ver
NFR-076	Every request that changes data, including Ask and its feedback, shall be refused with 403 and change nothing when its Origin header names another site or is "null" or malformed; a malformed Origin shall never cause a 5xx response.	T
NFR-077	A live group-room connection shall not outlive the session or the membership that opened it (FR-914); account deletion closes every connection of the account (FR-119).	T
NFR-078	No password, key or other credential shall be shipped in the app bundle, sample data or demonstration scripts.	T
NFR-079	Opening a link shall never by itself spend a student's question allowance or change the student's data; a tap by the student is required (FR-613).	I
NFR-080	The pre-push leak scan shall inspect every commit the remote does not yet have, including commits reached only by a new tag, and shall read files as committed, never through a local text converter.	T
NFR-081	A signed-in account shall be able to make at most 100 data-changing requests a minute (profile, Journey, groups, blocks); beyond that the server answers 429.	T
NFR-082	Limits keyed by network shall treat an IPv4 address, or an IPv6 /64 network, as one network.	T
NFR-083	The developer tools of Ask shall answer only on the developer's own machine with the developer switch on, and shall refuse a cross-site call even there.	T

Lead 9 of audit run-1 (the log-in limit is keyed per username, which the caller chooses) is **not** tightened per network on purpose: a school laboratory puts a whole class behind one network address. The network-wide limit of 100 a minute (NFR-037) still applies, and a human check (Cloudflare Turnstile, FR-912) is the planned fix. This is a recorded, accepted risk.

6 Business rules

Business rules are facts and policies of the admission domain that the system must respect. Their source is the UGC handbook and cut-off table for the 2025/26 intake; rules marked *ZedPath policy* are product decisions.

Sources are cited as **HB p.*n*** (printed page of the UGC handbook 2025/26 [2]; PDF page = printed + 7) and **COP** (UGC cut-off table 2025/26 [3]). Every handbook rule below was checked word for word against the PDF text. Rules marked **ZedPath policy** are product decisions; they are configurable and reviewed each intake.

6.1 General eligibility

Table 24: Business rules, general eligibility for state-university admission

ID	Rule	Source
BR-001	A candidate must have at least an *S* grade in each of three approved subjects in **one sitting**.	HB p.9
BR-002	A candidate must have at least **30%** in the Common General Paper (also called the Common General Test), in the current attempt or a previous one within the three allowed; without it the candidate cannot register even if selected.	HB p.9
BR-003	A candidate may use at most **three attempts**; after the third, results of any attempt cannot be used. Attempts need not be consecutive.	HB p.9, p.13
BR-004	Some candidates are ineligible regardless of results: those already registered as internal students of a UGC institution, degree holders, certain diploma and college students, holders of a government-channelled foreign scholarship, and anyone who made a false declaration.	HB p.13 to p.15
BR-005	A Z-score is entered with exactly four decimal places and must lie between −4.0000 and +4.0000. (Published cut-offs for 2025/26 range from −0.8505 to 2.7935.)	ZedPath policy; COP
BR-006	The candidate's district is the district of the school attended for the most days, totalling at least one year, in the three years before the examination; otherwise it is the district of permanent residence.	HB p.10, p.11
BR-007	Sixteen districts are educationally disadvantaged: Nuwara Eliya, Hambantota, Jaffna, Kilinochchi, Mannar, Mullaitivu, Vavuniya, Trincomalee, Batticaloa, Ampara, Puttalam, Anuradhapura, Polonnaruwa, Badulla, Monaragala and Ratnapura.	HB p.9

6.2 Streams, subjects and grades

Table 25: Business rules, streams, subjects and grades

ID	Rule	Source
BR-010	Subjects belong to six streams: Arts, Commerce, Biological Science, Physical Science, Engineering Technology and Biosystems Technology. A subject combination is admissible only if it follows the rules of its stream (for Arts, the basket rules 01 to 04).	HB p.20, p.31 to p.34, p.77, p.79
BR-011	Grades are ordered A > B > C > S > F. The handbook names S (Ordinary Pass), C (Credit Pass) and B (Very Good Pass); the full order is the Department of Examinations' grading scale.	HB p.9, p.83; ZedPath policy for the order

ID	Rule	Source
BR-012	The handbook gives no numeric subject codes; ZedPath assigns its own subject identifiers and maps every handbook subject name to one of them.	ZedPath policy

6.3 Selection and the preference list

Table 26: Business rules, quotas, selection and the preference list

ID	Rule	Source
BR-013	Up to 40% of places in a course are filled on all-island merit, up to 55% by district quota in proportion to population, and the remaining 5% are shared by the 16 educationally disadvantaged districts.	HB p.8, p.9
BR-014	Ten Arts courses (Arts, Arts SP, Arts SAB, Communication Studies, Peace and Conflict Resolution, Islamic Studies, Arabic Language, TESL, Social Work, Arts-IT) are filled **100% on all-island merit**; their cut-off is the same in every district (marked * in the COP).	HB p.8, p.21; COP
BR-015	Each course at each institution has a **Uni-Code**: a three-digit course code plus one institution letter (for example, 016C). There are 255 Uni-Codes for 2025/26.	HB p.112
BR-016	The UGC selects a candidate for the **highest-preferred Uni-Code** whose places are not already filled by candidates with higher Z-scores; a Uni-Code the candidate did not list is never allocated.	HB p.112, p.113
BR-017	A candidate may list at most **125** Uni-Codes and may not list the same Uni-Code twice.	HB p.113
BR-018	A candidate who does not register for a selected Uni-Code is not considered for the same course at any other university in later rounds, even if preferred higher.	HB p.114
BR-019	The order may be changed **once**, online, within two weeks of the closing date; the change replaces the whole list.	HB p.116, p.155

6.4 Course-specific requirements

Table 27: Business rules, course-specific requirements

ID	Rule	Source
BR-020	Each Uni-Code has subject and grade requirements that follow one of twenty patterns (fixed trio; trio with a count of higher grades; two fixed subjects plus a third from a list; one of several trios; slot rules; a gate subject at a higher grade; *n* of a list at a higher grade; and others). ZP-DOC-05 defines a rule grammar that expresses all twenty.	HB Section 2.2, p.31 to p.110
BR-021	Alternatives ("A or B") and combinations ("A and B") in a requirement are evaluated exactly as written; a requirement such as "a C grade in Combined Mathematics or Physics" is met by either subject.	HB p.70 (example: Computer Science)
BR-022	Fifty-one Uni-Codes also require G.C.E. (O/L) results (for example, a Credit Pass in English or Mathematics); some accept an A/L pass instead. The course text prevails over the summary table where they differ.	HB p.149, p.150; course pages

ID	Rule	Source
BR-023	Some requirements are not academic (Nursing: height at least 4 feet 10 inches and no deformity). They are shown as additional requirements and are not evaluated.	HB p.57, p.150
BR-024	Where a stream's "you may also apply for" list omits a course whose subject rule the student meets, the **subject rule prevails**.	ZedPath policy (handbook lists are incomplete)
BR-025	Twenty-four Uni-Codes require a practical or aptitude test run by the university; the student applies to the university directly after its press notice; passing does not remove the cut-off; failing affects only that course.	HB p.38, p.151
BR-026	Eight courses of the Gampaha Wickramarachchi University of Indigenous Medicine are suspended for 2025/26 and shall not be offered.	HB p.48, p.66, p.109
BR-027	Some courses are not taught in all three languages, so a student may meet the cut-off and still not be selected; this caveat is shown on every course page.	COP note; HB p.174

6.5 Cut-off data

Table 28: Business rules, cut-off data

ID	Rule	Source
BR-028	A cut-off is the minimum Z-score selected for one Uni-Code from one district in one intake year, published by the UGC after re-scrutiny. Previous cut-offs are guidance only and are presented as such.	HB p.10, p.112, p.173; COP
BR-029	*NQC* ("No Qualified Candidates") means no candidate from that district was selected for that course, for one of three stated reasons; it is not a numeric cut-off.	COP legend

6.6 Banding (ZedPath policy)

Bands turn cut-off history into an honest likelihood. They never promise admission (FR-207). Let Z be the student's Z-score (or the lower bound of an estimate), C the set of numeric cut-offs for the offering and the student's district over the most recent five intake years, n the number of values in C, and c_l the most recent value in C.

Table 29: Business rules, banding (ZedPath policy, configurable)

ID	Rule
BR-030	If $n = 0$ (every year NQC, or a new course), the band is **Not enough data**.
BR-031	If $n \geq 3$ the band uses *full history*; if n is 1 or 2 it uses *limited history* and is labelled "based on n year(s)". The margin m is 0.0000 for full history and 0.1000 for limited history.
BR-032	**Safe:** $Z \geq \max(C) + m$. **Likely:** $c_l \leq Z < \max(C) + m$. **Reach:** $\min(C) - 0.0500 \leq Z < c_l$. **Out of range:** $Z < \min(C) - 0.0500$. Comparisons use exact four-decimal values (NFR-051).
BR-033	Trend (full history only): the least-squares slope of C by year; **rising** above +0.0200 per year, **falling** below −0.0200, otherwise **steady**; **jumpy** if the standard deviation of C exceeds 0.1000 (overrides the others).
BR-034	An NQC year is excluded from C and shown as "No qualified candidates in *year*".
BR-035	For all-island-merit offerings (BR-014) the single all-island cut-off is used for every district.

ID	Rule
BR-036	The thresholds 0.0500, 0.1000 and 0.0200 are configuration values. When three or more years are loaded they shall be validated by back-testing: bands computed from earlier years are compared with the following year's actual cut-offs.
BR-037	*New (VR2).* Students see the bands as **Very likely** (Safe), **Possible** (Likely) and **Unlikely** (Reach), with these meanings: "Your Z-score is above every cut-off of the last five years"; "Above the latest cut-off, but not above every recent year"; "Below the latest cut-off: possible only if the cut-off falls". The codes SAFE, LIKELY and REACH and the rules BR-030 to BR-036 are unchanged.

6.7 Preference-list checks (ZedPath policy, derived from BR-016)

Table 30: Business rules, preference-list checks

ID	Rule
BR-040	Because selection takes the highest-listed Uni-Code the student clears (BR-016), an offering listed below one the student is very likely to clear is very unlikely to be reached.
BR-041	**Ordering warning:** when a Safe offering is listed above any Likely or Reach offering, the student is warned that the offerings below it are unlikely to be reached.
BR-042	**Coverage warning:** when the list contains no Safe offering, the student is warned that they may not be placed anywhere.
BR-043	The list enforces BR-017: at most 125 entries and no duplicates.

6.8 Accounts and contact details (ZedPath policy, CR-001 and CR-003)

New. The same rules are checked on the device, for a clear message, and on the server, which refuses anything the device let through. The database repeats them as constraints (ZP-DOC-05).

Table 31: Business rules, accounts and contact details

ID	Rule	Source
BR-050	A username has 3 to 24 characters from lower-case letters, digits, "_" and "."; it is stored and compared lower-cased and trimmed, and is unique.	ZedPath policy (CR-001)
BR-051	A password has 8 to 128 characters.	ZedPath policy (CR-001)
BR-052	A full name has 2 to 80 characters after trimming and joining spaces.	ZedPath policy (CR-003)
BR-053	An email address has the shape *name@domain.tld*, at most 254 characters, and is stored in lower case. It is not unique.	ZedPath policy (CR-003)
BR-054	A mobile number is a Sri Lankan mobile number, entered as 07..., +94 7..., 94 7... or 0094 7... with any spacing, and stored as +947 followed by eight digits. It is not unique (siblings may share a parent's number).	ZedPath policy (CR-003)
BR-055	A date of birth is a real calendar date, and the student's age on the day of sign-up is 13 to 70 years.	ZedPath policy (CR-003)
BR-056	A student under 18 confirms that a parent or guardian agrees to the details being kept.	ZedPath policy (CR-003); Personal Data Protection Act, No. 9 of 2022 [4]

ID	Rule	Source
BR-057	Consent is recorded with the version of the privacy wording agreed to (currently 2026-10-11) and the time; a new wording is a new version.	ZedPath policy (CR-003)
BR-058	A session lasts 30 days.	ZedPath policy (CR-001)
BR-059	*New (CR-007).* A demo account and its session last 24 hours from creation. Its username is "demo_" followed by eight random hexadecimal digits, and it has no usable password. It is deleted at the first demo tap after it expires (the free plan runs no scheduled clean-up). Demo taps share the log-in limits: 10 a minute per network address and 100 a minute per network address overall. *(CR-010)* At most 500 demo accounts exist at once, and each tap deletes at most 50 expired ones.	ZedPath policy (CR-007, CR-010)

6.9 Journey and reminders (ZedPath policy, CR-002, CR-006 and VR)

New.

Table 32: Business rules, Journey and reminders

ID	Rule	Source
BR-060	The Journey has eight steps: three preparation steps (enter results, build the list, compare courses) and five UGC steps (apply to the UGC, change the Uni-Code order once, apply for the aptitude test, selection results, register for the course). The aptitude step appears only when a course on the list has a test (BR-025).	ZedPath policy; HB p.116, p.118, p.151
BR-061	A preparation step can be ticked at any time. A UGC step can be ticked only on or after the opening date given in the UGC press notice, as a Sri Lanka date (UTC+05:30); while that date is not announced the step cannot be ticked. ZedPath never invents a date.	ZedPath policy (CR-006)
BR-062	*I applied* can be ticked on a followed notice only from the notice's opening date; a notice without an opening date is open from publication.	ZedPath policy (CR-006)
BR-063	Reminders fall 7 days and 1 day before a dated item, at 08:00 or 18:00 on the phone's clock; the default is both reminders at 08:00, the behaviour before settings existed.	ZedPath policy (CR-002, VR)
BR-064	*New (CR-009).* An aptitude-test notice is shown as the last intake's and *past* when the later of its closing date and test date is before today's Sri Lanka date, or when it states neither; the next intake's dates are never estimated. A course whose joint practical test is held at other universities shows those universities' notices.	ZedPath policy (CR-009); HB p.151 (BR-025)

6.10 Groups (ZedPath policy, CR-002, CR-005 and CR-006)

New.

Table 33: Business rules, groups

ID	Rule	Source
BR-070	There is at most one course group per course code and one batch group per pair of A/L stream and district. A group is created when its first member joins.	ZedPath policy (CR-002)
BR-071	Only a student whose results are saved on the account can join a batch group; it is the group of that student's stream and district.	ZedPath policy (CR-002)
BR-072	A message is text only, 1 to 500 characters. A member may post at most one message every 3 seconds and 20 in any 10 minutes in a group.	ZedPath policy (CR-002)
BR-073	A member may report a message once, for one of four reasons: harmful, wrong information, spam, other. A message reported by 3 different members is hidden for everyone.	ZedPath policy (CR-002)
BR-074	The group rules (version 2026-10-11) are six: be kind; stay on topic; never share phone numbers, email or home addresses, or passwords (ZedPath blocks phone numbers and emails); no selling, paid tuition adverts or exam leaks; members give opinions, so check official sources; report anything wrong (3 reports hide a message).	ZedPath policy (CR-005)
BR-075	The group rules are accepted once per account for each rules version, at the first join of any group; a new version must be accepted again, and earlier acceptances are kept. *Replaces the CR-005 rule "once per membership".*	ZedPath policy (CR-006)
BR-076	A senior invite names a course, an institution and a year of study from 1 to 6, is valid for 14 days, can be redeemed once, and only by a senior account of that institution. Its code is 80 random bits and is stored only as a SHA-256 hash.	ZedPath policy (CR-002)
BR-077	An avatar is one of 12 preset pictures (a01 to a12) or none, meaning initials. Photographs are never uploaded.	ZedPath policy (CR-006)
BR-078	*New (CR-010).* A report counts toward hiding a message (BR-073) only when the reporting account is at least one day old and is not a demo account.	ZedPath policy (CR-010, audit lead 4)
BR-079	*New (CR-010).* A member may make at most 20 post attempts in any 10 minutes in a group, counting refused ones; an account keeps at most three live connections to one group.	ZedPath policy (CR-010, audit leads 13 and 14)

6.11 Ask (ZedPath policy, VR)

New.

Table 34: Business rules, Ask

ID	Rule	Source
BR-080	ZedPath tells a student that the special intake for outstanding achievements **may** apply when they list a first, second or third place at national level, or an achievement at international level, dated 2023 to 2025. It only flags the possibility and cites the section; the handbook's further conditions (up to 0.5% of places per course, a Z-score within 0.2000 of the district cut-off) are decided by the UGC.	HB Section 6, p.166, p.167; ZedPath policy

6.12 University facts (CR-008)

New.

Table 35: Business rules, university facts

ID	Rule	Source
BR-081	University facts are matched to an offering by its institution letter (the last character of its Uni-Code). A university's Mahapola and Bursary totals include its campuses. Graduate figures for 2024 are provisional, and are marked incomplete when a faculty did not submit figures. "Information not submitted" means no value, never zero.	UGC, *Sri Lanka University Statistics 2024*, Tables 2.2, 3.1(a), 4.1, 7.1 and 7.2 [9]

6.13 Known gaps in the official sources

The 2025/26 handbook does not state: application opening and closing dates (announced by press notice), test dates for aptitude tests, a tie-break rule, a re-scrutiny procedure, Mahapola or bursary eligibility rules, hostel allocation rules, or the registration deadline. *Revised (CR-002, CR-006):* ZedPath shows these as *Not announced yet* with the official rule and its page (FR-402), never as confirmed facts and never as estimated dates. Subject means and standard deviations needed for the Z-score estimate (FR-107) are not in either source; FR-107 is therefore at risk until those statistics are obtained.

7 Verification

Each requirement states its verification method. ZP-DOC-08 maps every requirement to test cases and records results. A requirement is verified when its test cases pass, or when the inspection, analysis or demonstration is recorded with its evidence.

Appendix A: Traceability: stories to requirements

Table 36: Story-to-requirement traceability (every story in ZP-DOC-02 is covered)

Story	Title	Requirements
US-101	Enter my results once	FR-101, FR-102, FR-103, FR-117 (FR-104 superseded)
US-102	Change my profile at any time	FR-105, FR-106, FR-128, FR-130
US-103	Estimate my Z-score before results	FR-107, FR-108
US-104	Read my results sheet from a photo	FR-109, FR-110, FR-111
US-105	Use the app in my language	FR-112, FR-124
US-106	Create an account and use it on any device	FR-114, FR-115, FR-116, FR-117, FR-129, FR-131; NFR-036, NFR-037, NFR-082
US-107	Get back into my account with a recovery code	FR-118
US-108	Delete my account and everything in it	FR-119, FR-132; NFR-075
US-109	Give my contact details with a stated purpose and consent	FR-116, FR-120, FR-121; NFR-030, NFR-070
US-110	Choose a profile picture	FR-122
US-111	Find my subjects by searching	FR-123
US-112	Try ZedPath with a sample student	FR-125 to FR-128; NFR-078; BR-059
US-201	See every course I am eligible for	FR-113, FR-201, FR-202, FR-203
US-202	See why a course is hidden from me	FR-204
US-203	See Safe / Likely / Reach for each course	FR-205, FR-206, FR-207, FR-221; BR-037
US-204	Open a course page with sources	FR-208, FR-209, FR-218
US-205	Filter and sort courses	FR-210
US-206	Work backwards from a dream course	FR-211, FR-212
US-207	Compare up to three courses	FR-213, FR-216, FR-217
US-208	See what living there costs	FR-214
US-209	Find a course by searching	FR-215
US-210	See official facts about the university	FR-219; NFR-007, NFR-055; BR-081
US-211	See a course's aptitude test on its page	FR-220; BR-064
US-301	Build my preference list	FR-301, FR-302, FR-308
US-302	Get warned about a bad order	FR-303, FR-304, FR-305
US-303	Copy uni-codes in order	FR-306
US-304	Learn the UGC form step by step	FR-307
US-305	Select several courses at once	FR-309, FR-310, FR-311
US-401	See my journey with the next step first	FR-401
US-402	Know whether a date is confirmed or estimated	FR-402
US-403	Be told about aptitude tests for courses on my list	FR-403
US-404	Get reminders before deadlines	FR-404, FR-412 (FR-405, FR-406 superseded)
US-405	Check whether a special intake applies to me	FR-407
US-406	Add my own steps to the journey (superseded)	FR-408 (superseded by FR-414, FR-416)

Story	Title	Requirements
US-407	Tick my own Journey steps when they open	FR-409, FR-410; NFR-038
US-408	Choose how I am reminded	FR-413
US-409	Follow a course into my Journey	FR-416
US-410	Follow a notice or other path and record that I applied	FR-414, FR-415
US-411	Keep my Journey on my account	FR-411; NFR-075
US-501	See every route my results open	FR-501
US-502	Open a route's details	FR-502
US-503	Be alerted about job exams I qualify for	FR-503, FR-504, FR-509, FR-510, FR-417; NFR-072
US-504	Check whether a degree is recognised	FR-505
US-505	Check the interest-free loan scheme	FR-506
US-506	See scholarships abroad	FR-507
US-507	Understand what retrying would take (withdrawn)	FR-508 (withdrawn)
US-508	See how a vocational qualification leads to a degree	FR-511
US-601	Ask a question in my language	FR-601, FR-602, FR-613; NFR-079
US-602	Every answer cites its source	FR-603, FR-604
US-603	The assistant won't predict the future	FR-605
US-604	Turn an answer into an action	FR-606
US-605	Read a clear answer card	FR-607, FR-608, FR-609, FR-610
US-606	Tell ZedPath whether an answer helped	FR-611, FR-614; NFR-073
US-607	Let Ask use my achievements and interests	FR-612; NFR-074; BR-080
US-701	Share a summary with my family	FR-701, FR-702
US-702	Ask a verified senior	FR-703
US-703	Become a verified senior	FR-707 (FR-704 superseded)
US-704	Join a group for my batch and target course	FR-705, FR-715, FR-716, FR-719; UI-4; NFR-043
US-705	See my class's progress as a teacher	FR-706
US-706	Stay safe in groups	FR-708, FR-709, FR-710, FR-711, FR-712, FR-720 to FR-723; NFR-039; BR-078, BR-079
US-707	Accept the group rules once	FR-714 (FR-713 superseded); NFR-038
US-708	See a group's details and members, and leave it	FR-717, FR-718; NFR-071
US-801	Get my registration checklist	FR-801
US-802	Check Mahapola and bursary	FR-802
US-803	Plan the gap year	FR-803
US-804	Keep my documents ready	FR-804
US-901	Register an official source	FR-901, FR-902
US-902	Extract rules from the handbook with AI	FR-903, FR-904, FR-905
US-903	Review uncertain facts	FR-906, FR-907
US-904	Scan the Gazette every week	FR-908, FR-909
US-905	Report a mistake	FR-910
US-906	Keep a history of every change	FR-911

Story	Title	Requirements
US-907	Protect the app from abuse	FR-912, FR-913, FR-615; NFR-054, NFR-080 to NFR-083
US-908	Close my open chats when my session ends	FR-914; NFR-076, NFR-077

Appendix B: Glossary

Table 37: Terms and abbreviations used in this document

Term	Meaning
Band	ZedPath's classification of an offering relative to the student's Z-score: Safe, Likely, Reach, Out of range, Not enough data
COP	The UGC table of minimum Z-scores ("cut-off points") by course, university and district
Demo account	A private, throwaway account holding the sample student, opened by *Try the demo* and deleted after 24 hours (CR-007)
ETag	An HTTP response header naming a version of a response; a client sends it back so the server can answer 304 Not Modified
Fingerprint	A SHA-256 hash of a normalised web address, used to block a page without publishing its address
NQC	"No Qualified Candidates": a cut-off cell for which no candidate from that district was selected
NVQ	National Vocational Qualification, Levels 1 to 7 (Tertiary and Vocational Education Commission)
Origin	An HTTP request header naming the website a request came from; ZedPath refuses data-changing requests from other sites
Offering	One course of study at one university
PBKDF2	Password-Based Key Derivation Function 2: stretches a password into a key with many hash rounds
SRS	Software Requirements Specification
VAPID	Voluntary Application Server Identification for Web Push (RFC 8292)
Verdict	The one-word answer shown above an Ask answer: Yes, No, It depends or Not in official sources (CR-004)
VR	The product owner's video review of the live app, 11 October 2026
VR2	The product owner's review and decisions of 14 October 2026 (band names, intro guide, avatars, expired session)
Window	The period in which a Journey step or *I applied* can be ticked (BR-061, BR-062)

Appendix C: References

Table 38: Sources referred to in this document

Ref	Source
[1]	ISO/IEC/IEEE 29148:2018, *Requirements engineering*
[2]	University Grants Commission, *Admission to Undergraduate Courses of the Universities in Sri Lanka, Academic Year 2025/2026* (student handbook)
[3]	University Grants Commission, *Minimum "Z" Scores for selection to various Courses of Study, Academic Year 2025/2026*, 31 July 2026
[4]	Personal Data Protection Act, No. 9 of 2022, Democratic Socialist Republic of Sri Lanka
[5]	W3C, *Web Content Accessibility Guidelines (WCAG) 2.2*
[6]	IETF RFC 5545 (iCalendar), RFC 8030 (Web Push), RFC 8292 (VAPID)
[7]	ZP-DOC-02 User Stories; ZP-DOC-06 Software Architecture
[8]	ZP-DOC-05 Data Design, sections on CR-001 to CR-007 (migrations 0003 and 0005 to 0011) and the university-facts read model
[9]	University Grants Commission, *Sri Lanka University Statistics 2024*, Chapters 2, 3, 4 and 7
[10]	Tertiary and Vocational Education Commission, *NVQ Framework of Sri Lanka: Operations Manual*, October 2009
[11]	ZP-DOC-08 Test Plan and Test Summary Report (test-campaign findings D-01 to D-13; security audit run-1)